ii) Ultra-violet radiation

The ultraviolet (UV) portion of the electromagnetic spectrum includes all radiations from 150-3900A°, UV radiation around 2600A° is most lethal to microorganisms. UV has a very little ability to penetrate matter. Thus, only the microorganisms on the surface of an object, exposed directly to the ultraviolet light are susceptible to destruction. UV radiations are used to sterilize operation theaters, laboratories and entry ways.

b) Ionizing radiation

Ionizing radiations (X-rays, Gamma rays and Cosmic rays) are an excellent sterilizing agents and they penetrate deep into the objects. These radiations do not produce heat on the surface of materials. Hence, sterilization using ionizing radiations is referred as cold sterilization. It will destroy bacterial endospores and vegetative cells, both Prokaryotic and Eukaryotic; however ionizing radiation is not always effective against viruses. Gamma radiation from Cobalt 60 source is used in the cold sterilization of antibiotics, hormones, sutures and plastic disposables supplied such as syringes and in pasteurization of meat.



Deinococcus radiodurans is an extremophilic bacterium. It is one of the most radiation-resistant organisms known.

4.6 Filtration

Filtration is an effective and reasonably economical method of sterilization. It is used to sterilize heat-sensitive fluids, and air. It is particularly useful for solutions containing toxins, enzymes, drug, serum and sugars. Sugar solutions used for the cultivation of microorganisms tend to caramelise during autoclaving and so they are best sterilized by filtration. Filtration is also used extensively in beer and wine industries. Filters with known pore sizes which are sufficiently small to hold back bacteria are employed. Recently filters that can remove viruses are also available.

Filtration is an excellent way to remove the microbial population from solution containing heat sensitive material.

There are two types of filters namely (Figure 4.4):

i) Membrane filter (surface filtration) and

ii) Depth filter

Membrane filters

Membrane filtration is used for preparing heat-labile culture media components. It

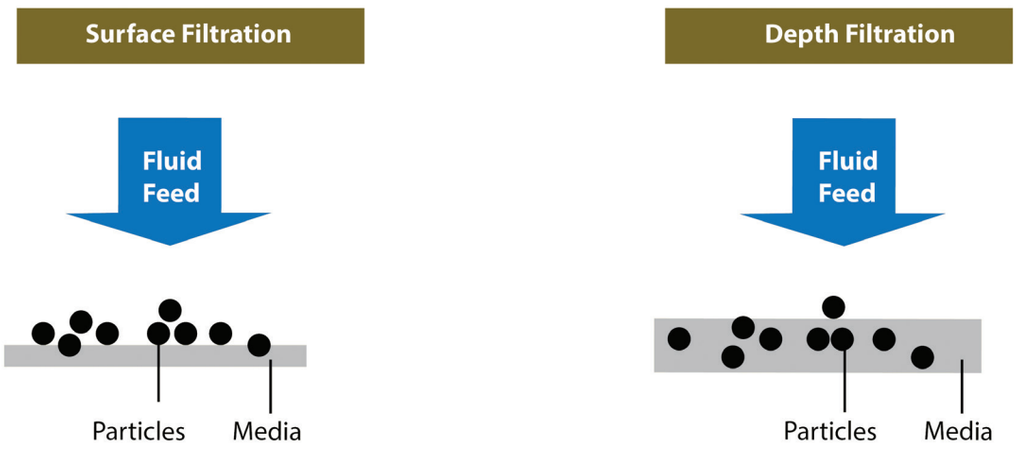


Figure 4.4: Principle of filtration

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